

Module 4 — Maneuvers & Perturbations (Orbit Academy v5.5)

Printable edition. The interactive version — live simulator, self-checking questions — is the course website; open the matching tut/worksheet pages there. Dynamic diagrams appear here as bracketed figure notes.

What you will learn

- What **delta-v** (written Δv) is — a change in speed that you spend to reshape an orbit — and why a satellite carries only a **finite supply**.
- Why **tangential (prograde/retrograde) burns** reshape an orbit efficiently while **radial burns** mostly waste fuel — including the paradox that pushing “up” can drop you into the Earth.
- How every big orbit change is really a **two-step maneuver** — raise your apogee, then match circular speed — as in a **LEO→GTO→GEO** transfer.
- What **escape velocity** is (~ 11.2 km/s at the surface) and how a fast-enough burn opens a closed orbit into an **unbound** path.
- How **atmospheric drag** decays a low orbit toward re-entry — and the surprise that a satellite steadily *losing* energy to drag ends up moving **faster**.
- What a **perturbation** is, and why continuous forces (drag, J2, sunlight, ion thrust) make orbits — and TLEs — **go stale**.

Quick-start tutorial

A perfect Keplerian orbit assumes only a point-mass Earth. In the real world, extra forces nudge a satellite off that ideal ellipse. Any such nudge — from air, sunlight, Earth’s bulge, or a thruster — is a **perturbation**: a small push that changes the orbit over time.

To change an orbit on purpose, you fire a thruster. The size of that shove is measured as **delta-v** — literally a **change in speed**, in m/s or km/s. It is the true currency of spaceflight: every satellite launches with a fixed tank of propellant, and Δv is how we count what is left. When the Δv budget runs out the mission is over, even if every other system still works.

Picture this: Delta-v is the cash in your wallet. Every maneuver has a price tag, you pay it once, and you can never earn it back. Spend wisely — a wasteful burn is money you’ll wish you had at end of mission.

Where you point the burn matters enormously

A **tangential** burn (prograde = speed up, retrograde = slow down) points along your direction of travel. It does real work on the orbit and efficiently raises or lowers the **opposite** side — that “a burn changes the opposite side” rule belongs to along-track burns. A **radial** burn (straight toward or away from Earth) does almost no work — it mostly **rotates** the ellipse. Shove hard “up” and a new, much lower perigee appears a quarter of the ring away from you; push hard enough and it can drop into the Earth.

[Figure: A prograde burn (arrow) leaves your burn point as the low point and lifts the far side into a taller ellipse — a higher apogee.]

Big moves take two steps

To reach a much higher orbit you (1) burn **prograde** to raise your apogee out to the target distance, then (2) at that apogee, burn again to **match the circular speed** there. That's the LEO→GTO→GEO transfer. Burn hard enough (about **11.2 km/s** from the surface, less from orbit) and you reach **escape velocity**: the orbit opens up and you never come back.

Finally, in low orbits the thin upper air causes **drag**, shrinking the orbit until re-entry (~7.8 km/s, about 17,500 mph). The twist — and it is worth stating carefully, because the short version is simply wrong: **drag never pushes a satellite along**. It only ever takes energy away. But losing energy makes the orbit *smaller*, and a smaller orbit demands a *higher* speed, so the satellite ends up moving faster while being slowed the whole way down. Because drag, sunlight, and Earth's bulge act continuously, a satellite's orbit is always drifting — which is why a TLE snapshot goes stale.

The forces that perturb an orbit

A Keplerian orbit assumes a perfect point-mass Earth and nothing else. These are the real-world forces that deflect a satellite from that ideal ellipse — plus the two engines used to fight back or reshape the orbit on purpose.

What it sets	Plain-language meaning	In the tool
Atmospheric drag	Thin upper-atmosphere gas pushes against a satellite, sapping energy and shrinking the orbit. Dominant in LEO; negligible above ~1,000 km. Ends in re-entry (burn-up or break-up).	"Drag decay" scenario — watch a low orbit spiral in.
Earth's equatorial bulge ("J2")	Earth isn't a perfect sphere; the bulge exerts a torque that slowly swivels the orbit plane (its RAAN precesses). Used on purpose for sun-synchronous orbits. ("J2" is just the label physicists give this bulge term when they write Earth's gravity as a series of numbered pieces — J2 is the biggest one after a perfect sphere.)	"Sun-synchronous" scenario — watch the plane precess.
Solar radiation pressure	Sunlight is a "photon wind" that pushes on the satellite. Effect scales with illuminated area ÷ mass, so light, high-area objects feel it most. Matters above ~1,000 km.	(described here — this sim models drag only, below 1,000 km; the same area-to-mass slider governs both)
Outgassing	Trapped gases venting from the spacecraft give tiny, unpredictable nudges.	(described here; too small to simulate)
Chemical thrusters	Burns propellant for a short, powerful, "impulsive" shove. Orbit jumps from one ellipse to another, then coasts predictably.	The burn buttons — one press = one impulse.

What it sets	Plain-language meaning	In the tool
Electric (ion) thruster	Uses solar-cell power for a gentle, continuous thrust over days/weeks. Great efficiency, but the orbit is always changing — hard to predict (e.g. Starlink).	(described here; contrasts with impulsive burns)

PART A • Why prediction breaks down

The limits of the tidy six-number picture, and the two kinds of engine.

Exercise 1 — The limits of a TLE (and continuously-thrusting satellites)

In the last module you learned to read a **TLE** — a tidy little packet of six numbers that pins down one exact ellipse at one exact instant. It’s a snapshot. And like any snapshot, it’s frozen: it captures where the satellite was and where it was headed *at that moment*, on the assumption that nothing but plain Earth gravity is acting on it.

The trouble is that the real world never leaves an orbit alone. Air, sunlight, Earth’s own lumpy shape, and the satellite’s own thrusters all keep nudging it off that ideal ellipse. So the snapshot starts drifting out of date the instant it’s taken — and that “going stale” is the thread running through this entire module.

There’s one case where the snapshot fails almost immediately: a satellite that is **thrusting continuously**. If a spacecraft is gently pushing the whole time (as the **Starlink** satellites do with their electric thrusters), it’s never actually *on* any single fixed ellipse — so no single TLE can honestly describe it for long.

[Figure/insert: see the online worksheet]

Try it in the simulator:

Recall from the last module that a TLE is a snapshot of one ideal ellipse at one instant. Read the propulsion note below, then consider where that promise breaks.

What to look for: a TLE only ever describes one frozen ellipse — so anything that keeps changing the orbit (a perturbation, or a continuously-firing thruster) is exactly what a TLE can’t keep up with.

While you watch, think about:

- A TLE stores the inclination and eccentricity just fine — so what is it fundamentally *unable* to capture?
- Why would a satellite that fires its thruster for weeks on end be almost impossible to pin to a single TLE?
- If you had to track a Starlink that’s slowly raising its orbit, how often do you think you’d need a fresh snapshot?

Check your understanding: What is a fundamental limitation of representing an orbit as a TLE (six fixed elements at one epoch)?

- It can only describe circular orbits, so any real ellipse has to be approximated
- It captures one unperturbed ellipse, so it shows neither perturbations nor thrust
- It has no field for inclination, so the plane’s tilt must be supplied separately

- It only works above the GEO belt, where the atmosphere no longer interferes

Exercise 2 — Impulsive vs. continuous thrust

There are two very different ways to push a spacecraft, and the difference decides how predictable it is. A **chemical thruster** gives a short, hard kick — a burn lasting seconds or minutes — and then the engine is off. Before and after that kick, the satellite simply **coasts on gravity alone**, tracing a clean, predictable ellipse. We call such a brief, powerful shove an **impulsive** burn.

An **electric (ion) thruster** works the opposite way. It sips electricity from solar panels and puts out a whisper of thrust — but it keeps pushing for *days or weeks without stopping*. That's wonderfully fuel-efficient, but it means the orbit is **always changing**: there's never a long stretch of pure coasting to lock a prediction onto. A non-gravity force is acting the entire time.

So a big chemical satellite way out in high orbit is easy to forecast — long, quiet coasts between rare burns. A continuously-thrusting ion craft like a **Starlink** is genuinely hard to forecast, not because it's hiding, but because it never holds still on one orbit.

The point: a chemical burn is brief — the satellite coasts predictably before and after. An ion thruster pushes the whole time, so the orbit is never on a single fixed ellipse.

While you watch, think about:

- Between two chemical burns, what is the satellite doing — and why does that make it easy to predict?
- Why does an efficiency win (the gentle ion thruster) come with a prediction cost?
- Which would you rather track with a once-a-day observation: a chemical satellite in high orbit, or a thrusting Starlink? Why?

Check your understanding: Why is a big chemical-thruster satellite in high orbit easier to predict than an ion-powered Starlink?

- Chemical satellites fly high enough that gravity barely bends their path at all
- The chemical one coasts between brief burns; the ion one thrusts continuously
- Starlinks are physically too small for ground radar to track with precision
- Ion thrusters are far more powerful, so the errors they introduce are larger

PART B · Delta-v: the currency of maneuvers

Every orbit change costs a change in speed. Learn to spend it wisely.

Exercise 3 — Delta-v and prograde/retrograde burns

To change an orbit, you fire a thruster, and what you physically spend is **propellant** — the tank really does empty. Engineers nevertheless budget a mission in **delta-v** (Δv), the change in speed that propellant buys, measured in m/s or km/s. The reason is that the Δv a maneuver costs is a property of the *maneuver*, the same for a cubesat and a space station, while the propellant it takes depends on how heavy your spacecraft is and how good its engine is. Think of it as the **cash in the satellite's wallet**: it launches with a fixed tank of delta-v, every maneuver has a price, you pay it once, and you can never

earn it back. When the delta-v runs out, the mission is over, even if every other system still works.

Where you point that burn matters. A **prograde** burn fires along your direction of travel and **speeds you up**; a **retrograde** burn fires backward and **slows you down**. Here's the part that trips people up: the burn doesn't move the point you're standing at. Instead, it changes the **opposite side** of the orbit. Speed up, and the far side climbs higher; slow down, and the far side drops.

Predict first: Before you fire: you're in a circular orbit and you burn **prograde** (speed up) at one point. Do you think the whole ring grows evenly, or does one particular side of the orbit move? If one side moves, which one — the point you burned at, or the far side?

Try it in the simulator:

1. Open the simulator, drag the **starting circular altitude** slider up to ~1,500 km (high enough that no air muddies the picture — the tool opens in the low, draggy Drag-decay orbit) and click ► **place satellite in circular orbit**. Set a burn of ~200 m/s and click ↑ **prograde (speed up)**. Watch what happens to apogee and perigee.
2. Click ► **place satellite in circular orbit** again to start fresh, then apply the same ~200 m/s burn ↓ **retrograde (slow down)** and compare which side of the orbit moves this time.
3. Fire a couple more burns in a row and watch the □ **Δv budget gauge** at the top of the readout panel tick down each time — notice you can't get that fuel back.

What to look for: a prograde burn raises the far side of the orbit (apogee climbs); a retrograde burn lowers it. Your burn point stays put — the opposite side moves. Each burn draws down the fuel gauge: a satellite launches with a fixed tank of Δv , and when it's empty, the mission is over.

While you watch, think about:

- After a prograde burn, which point of the new orbit is your burn location — the high point or the low point?
- Did the gauge refund any fuel when you burned the other direction, or did every burn cost you?
- If your whole mission had to fit inside one small tank of Δv , how would that change the way you'd plan maneuvers?

Check your understanding: You are in a circular orbit and fire a prograde (speed-up) burn. What happens?

- The whole orbit expands uniformly outward, staying a slightly larger circle
- Your burn point becomes perigee and the opposite side rises — apogee climbs
- The orbit shrinks, since adding speed pulls you into a tighter, faster path
- Nothing changes until you burn a second time to close the new orbit off

Exercise 4 — Radial vs. tangential: which holds an orbit up?

A burn can point two very different ways. A **tangential** burn fires along your direction of travel (prograde or retrograde) — the same direction you're already moving. A **radial** burn fires straight toward or straight away from Earth, sideways to your motion. It seems like pushing "up," away from Earth, ought to be the natural way to climb to a higher orbit. It isn't.

The reason comes down to **work**. A burn adds energy to your orbit most effectively when it pushes in the same direction you're already going — so a tangential (prograde) burn does real work and efficiently lifts the orbit. A radial burn is aimed roughly *across* your motion, so it does almost no work on the orbit's energy. It mostly just tips and reshapes the ellipse without making it meaningfully bigger. To raise an orbit, burn tangential; radial thrust is a fuel-waster.

Predict first: Before you burn: you have a fixed amount of Δv to spend. Which do you think will lift your orbit more — a **radial** burn pointed straight away from Earth, or a **tangential prograde** burn along your direction of travel?

Try it in the simulator:

1. Open the **Station-keeping** scenario. With a fixed Δv , apply a **radial** burn (toward/away from Earth) and note how much apogee and perigee actually change.
2. Reset and spend that same Δv on a **tangential prograde** burn instead, and compare — watch which one genuinely raises the orbit and which just reshapes it.

What to look for: the tangential (prograde) burn efficiently raises the orbit; the radial burn mostly just rotates/reshapes the ellipse and barely changes its size. Tangential thrust does real work on the orbit's energy; radial thrust does almost none.

While you watch, think about:

- For the same fuel spent, which burn direction gave you the bigger, higher orbit?
- The radial burn changed the orbit's *shape* but not its size much — where did all that fuel's energy go?
- Why does a burn “along your motion” do more useful work than one aimed across it?

Check your understanding: To efficiently raise a decaying orbit, in which direction should you burn?

- Radially outward, straight away from Earth toward the altitude you want
- Tangentially, prograde — along the direction you are already travelling
- Radially inward, using the rebound to carry you out the far side
- Direction makes no difference; only the size of the burn matters at all

Exercise 5 — What if you kill the tangential speed entirely?

Back in Module 1 you learned the one-sentence definition of an orbit: it's **falling sideways fast enough to keep missing the ground**. The whole trick is the sideways speed — the ground curves away beneath you exactly as fast as you fall, so you never quite hit it.

So here's a sharp question. What if you took away that sideways speed entirely? Fire a retrograde burn big enough to cancel *all* of your tangential motion, bringing it to a dead stop relative to the ground below. There's nothing left carrying you “around” the Earth. You're not in an orbit anymore — you're just an object hanging in space with gravity pulling on it. And gravity always wins that argument: you fall straight down.

Predict first: Before you try it: if you cancel your sideways speed completely, do you think you'll hover in place at that altitude, drift gently outward, or fall straight toward Earth?

Try it in the simulator:

Thought experiment (you can approximate it: in a circular orbit, apply a big retrograde burn equal to your speed). Imagine a burn that brings the sideways (tangential) speed all the way to zero, then watch which way the object goes.

What to look for: with no sideways motion, nothing carries you “around” the Earth anymore — there’s no orbit left. You simply fall.

While you watch, think about:

- Once the sideways speed is gone, what force is left acting on the object — and which way does it point?
- Why can’t the object just “hover” at its altitude with zero sideways speed?
- How is this the same idea as an orbit, just with the one ingredient (sideways speed) removed?

Check your understanding: From a circular orbit, you make a burn that brings your tangential (sideways) speed to exactly zero. What happens?

- You stay at the same altitude, simply hanging there without moving
- You fall straight down, radially inward, all the way toward the Earth
- You drift slowly outward, since gravity weakens as the orbit relaxes
- You enter a new circular orbit, smaller but still perfectly closed

Exercise 6 — The radial-burn paradox: pushing “up” crashes you down

This is one of the most counterintuitive results in all of orbital mechanics, so it’s worth meeting head-on. Your gut says: to get higher, push up. Fire your thruster straight away from Earth and you should climb to a higher orbit. It feels obvious. It’s also wrong — and dangerously so.

Remember what you just learned: a radial burn does **almost no work** on the orbit’s energy. (The handy “a burn changes the opposite side” rule belongs to **along-track** burns — a radial burn does something sneakier.) Pushing “up” **rotates the ellipse** about a quarter-turn: your burn point stays at about the same distance — and so does the point directly opposite you — but a new, much lower **perigee** appears a quarter of the ring away. Push hard enough and that low point drops *below the surface* — so before the lap is out, the satellite you tried to lift comes plunging into the Earth.

Predict first: Before you fire: you’re in a 1,500 km circular orbit and you shove hard **straight up**, away from Earth. Do you think you’ll climb into a higher circular orbit — or could pushing “up” actually send you crashing down? Where do you predict the new low point of the orbit will end up?

Try it in the simulator:

1. Open the **Radial-burn paradox** scenario (a 1,500 km circular orbit). Set a burn of about **2,000 m/s** and press **radial out** — straight away from Earth.
2. Watch the **perigee** readout and the altitude-vs-time plot as the orbit evolves.
3. For contrast, reset and spend a similar Δv **prograde** instead, and compare: the prograde burn lifts the far side, the radial burn drops it.

What to look for: pushing straight “up” does *not* raise your orbit. It leaves your burn point at roughly the same distance and rotates the ellipse, planting a much lower perigee

a quarter of the ring away — below the surface — so before the lap is out the satellite plunges into the Earth.

While you watch, think about:

- You pushed away from Earth, yet a point elsewhere on the ring swung *toward* Earth. Why does the burn reshape the orbit somewhere else, instead of lifting the point you burned at?
- How does the radial result compare to spending the same fuel on a prograde burn?
- What does this tell you about trusting your everyday “push up to go up” intuition in orbit?

Check your understanding: From a circular orbit, you make a large burn pointed radially outward (straight away from Earth). What happens?

- You climb smoothly into a higher circular orbit centred on the same point
- The orbit goes lopsided and perigee drops sharply — hard enough, below the surface
- Nothing happens, because a purely radial burn cannot change an orbit at all
- You escape Earth entirely, since the thrust points directly outward from it

PART C · Get from LEO to GEO — the challenge

First try it your own way and tally the fuel. Then discover the efficient two-step recipe.

Exercise 7 — Challenge: get from LEO to GEO your own way

Here’s a real mission problem: you’re in a **circular low orbit** (~400 km) and you need to reach a **circular orbit at GEO** (~35,800 km up). You have a thruster and a limited tank of **delta-v**. How little can you spend?

Before anyone tells you the “right” way, **just try it**. Fire burns in whatever directions you like — toward Earth, away from Earth, along your motion, against it — and watch the orbit change. The **Δv gauge** keeps a running total of everything you spend. Getting to GEO is the goal; getting there *cheaply* is the real game.

Predict first: Before you touch anything: to climb from a low orbit way out to GEO, do you think it’s more efficient to push **outward** (straight away from Earth, “toward” GEO), or to push **along your direction of travel** (speeding up)?

Try it in the simulator:

1. Open the **GTO transfer** scenario (it starts you in a 400 km circular LEO). Note the **Δv gauge** reads 0.
2. Try to reach GEO however you like — experiment with **radial-out** (↖, straight away from Earth) and **prograde** (→, along your motion) burns. Watch the apogee/perigee readouts and the running **Δv** total.
3. When you’ve either reached GEO or given up, read your **total Δv spent**. Then hit **↺ refuel / reset budget** (that only refills the tank — it doesn’t move the satellite) and re-click the **GTO transfer** scenario to start back at 400 km, and try a different strategy — can you do it for less?

What to look for: burns straight “out” toward GEO barely raise your orbit for the fuel spent, while burns along your motion (prograde) lift the far side of the orbit dramatically — so how you point the thruster matters enormously.

While you watch, think about:

- Which burns moved you toward GEO the most per unit of Δv — the ones pointing outward, or the ones along your motion?
- When you pushed straight “out,” what actually happened to the orbit’s shape?
- What was your best (lowest) total Δv , and do you think you could still do better?

Check your understanding: After experimenting, which kind of burn moved you toward GEO most efficiently (most orbit-raising per unit of Δv)?

- Radial burns, pushing straight out toward the altitude you are aiming for
- Burns along your direction of travel — prograde, speeding you up
- Burns against your motion, retrograde, letting gravity do the lifting
- Direction made no measurable difference to the fuel each attempt cost

Exercise 8 — The efficient way: raise apogee, then circularize

Here’s the efficient recipe the pros use — the **two-step transfer**. Do it in two well-aimed burns, both **along your direction of travel**:

Step 1 — raise your apogee to GEO. One prograde burn at your low orbit of about **2,400 m/s** stretches your circular orbit into a tall ellipse whose far point (apogee) just reaches GEO altitude. That ellipse is the **Geo-Transfer Orbit (GTO)**. Now coast — for free — all the way out to apogee. **Step 2 — circularize.** Way out at apogee you’re moving too slowly to hold a circle there (~ 1.6 km/s vs. the ~ 3.1 km/s a GEO circle needs), so a second prograde burn of about **1,460 m/s** speeds you up and rounds the orbit off at GEO.

Total: about **3,900 m/s**. Compare that to whatever you spent flailing in the last exercise — this is almost always far less.

And here is why this recipe is worth memorizing: it is **universal**. Raise your high point to where you want it, then match the circular speed once you arrive — LEO→GEO, a small station-keeping raise, even a shot out to the Moon. The scale changes; the two steps never do.

Predict first: The efficient transfer takes two burns. Where along the orbit do you think each burn should happen — and which direction should each point?

Try it in the simulator:

1. Hit **↺ refuel / reset budget**, then re-click the **GTO transfer** scenario to get back on the 400 km circular orbit (refuel alone doesn’t reset the orbit). Fire a **prograde** burn at your low orbit and tune it ($\sim 2,400$ m/s) until the **apogee readout reaches GEO altitude** ($\sim 35,800$ km).
2. Speed up time and **coast** out to apogee (don’t burn during the coast — it’s free).
3. At apogee, fire a second **prograde** burn ($\sim 1,460$ m/s) until the orbit circularizes — watch for the “**□ Circular GEO reached**” banner. Read your total Δv ($\sim 3,900$ m/s).

What to look for: two clean prograde burns — one to raise apogee, one to circularize — reach GEO for about 3,900 m/s total, typically far less than free-form experimenting.

While you watch, think about:

- How does your two-step total ($\sim 3,900$ m/s) compare with your best free-form attempt?

- Why is it worth coasting all the way to apogee before the second burn, instead of burning early?
- Both burns point the same way (prograde). Why is neither one radial?

Check your understanding: In a GTO transfer, why is the second (circularizing) burn done at apogee — the highest point?

- Because thrusters only reach full efficiency far from Earth's atmosphere
- Because apogee already sits at GEO, so a burn there lifts perigee to match it
- Because the satellite moves fastest at apogee, making the burn most effective
- Because gravity is strongest there, and burning against it saves the most fuel

Exercise 9 — Which burn sequence actually reaches circular GEO?

You've got the recipe — now let's pressure-test it. Below are four plausible-sounding plans for getting from a circular 400 km LEO to a circular GEO. Only one actually works, and the wrong three each fail in an instructive way you can now predict.

Before you check the sim, reason each one through with what you know: one burn only moves the opposite side of the orbit; radial burns do almost no work; and a retrograde burn at apogee *lowers* perigee rather than raising it. Then run them in the simulator and watch each candidate either round out at GEO — or betray its flaw.

[Figure/insert: see the online worksheet]

Predict first: Before you test: read the four sequences below and place your bet. Which single sequence do you think delivers a truly **circular** orbit at GEO — and can you already name why each of the other three falls short?

Try it in the simulator:

Think through (and test in the sim) these candidate procedures for getting from a 400 km circular LEO to a circular GEO orbit. Run each one and watch whether it ends round at GEO or reveals its flaw.

What to look for: only the two-burn sequence (B) leaves you round at GEO — the others each fall back to LEO, barely move, or drop perigee even lower.

While you watch, think about:

- For Sequence A, where is perigee after the single burn — and what does that mean the satellite does twice per orbit?
- Sequence D uses the right timing (a burn at apogee) but the wrong direction. Why does retrograde there make things worse, not better?
- What do the three failures have in common with the mistakes you saw in Part B?

Check your understanding: Which sequence correctly delivers a circular orbit at GEO?

- Sequence A — a single large prograde burn while still down at LEO
- Sequence B — raise apogee to GEO, coast, then burn prograde at apogee
- Sequence C — a radial-outward burn held until you drift up to the belt
- Sequence D — raise apogee to GEO, coast, then burn retrograde at apogee

PART D · Escape velocity, unbound orbits & infall

Burn hard enough and you never come back; cancel your sideways speed and you fall straight in.

Exercise 10 — Escape velocity

So far every burn has left you **bound** — going around and around on a closed loop, always coming back. But there's a threshold. Push your speed high enough and you gain more energy than Earth's gravity can ever claw back, and you coast away for good. That threshold speed is **escape velocity**.

The number is worth carrying in your head: about **11.2 km/s** right at Earth's surface — roughly 25,000 mph. It isn't a magic constant; it's always exactly $\sqrt{2}$ (**about 1.41**) **times the local circular speed**. And because gravity is weaker the higher you go, escape velocity *drops* with altitude — from the 1,500 km orbit in this scenario it's only about 10.1 km/s. Below escape speed you stay in a closed loop; reach it and the loop springs open.

[Figure/insert: see the online worksheet]

Predict first: Before you check the readout: how fast do you think you have to be going to leave Earth for good from the surface — a few km/s, around 11 km/s, or something enormous like 100 km/s?

Try it in the simulator:

Open the **Escape & unbound** scenario. Note the **escape speed** readout in the right panel and compare it to your current orbital speed — see how much more you'd need to add. Then imagine (or try) nudging your altitude higher and watch the escape-speed figure drop.

What to look for: escape speed is only about $1.4\times$ your circular speed — closer than you might expect — and it gets smaller the higher up you already are.

While you watch, think about:

- How much faster than your current circular speed do you actually need to go to escape — a little, or a lot?
- Why does escape velocity get *smaller* the farther you are from Earth?
- If escape is $\sqrt{2}$ times circular speed, roughly what would escape speed be from an orbit where circular speed is 7 km/s?

Check your understanding: Roughly what is escape velocity from the surface of the Earth?

- About 1 km/s, a little faster than a rifle bullet leaves the barrel
- About 11 km/s, which is roughly 25,000 mph
- About 100 km/s, comparable to the speed of the solar wind
- About 300,000 km/s, the same as the speed of light in vacuum

Exercise 11 — Crossing escape: the orbit opens into a hyperbola

What actually happens to the *shape* of an orbit as you cross the escape threshold? The answer is a clean, dramatic switch. A bound orbit is a closed loop — a circle or an ellipse

— with an **eccentricity below 1.0**. As you add speed, that ellipse stretches longer and longer, its eccentricity creeping up toward 1.

At exactly escape speed, eccentricity hits **1.0** and the loop stops closing — it becomes an open **parabola**. Push past that and eccentricity climbs above 1, giving a **hyperbola**: a one-way curve that swings past Earth once and heads out, never to return. That open curve isn't just a spaceflight thing — it's the exact geometry of a comet or spacecraft **flying by** a planet, or any object "scattering" past a body it doesn't quite get captured by.

Predict first: Before you burn: you've watched faster burns make bigger and bigger ellipses that always come back. Do you think there's a speed past which the orbit stops being a closed loop entirely? What do you predict the eccentricity readout will do as you cross escape speed?

Try it in the simulator:

1. In the **Escape & unbound** scenario, escape with a **prograde** burn ($\sim 2,950$ m/s) and watch the eccentricity readout climb past 1.0 as the orbit opens.
2. Reset and escape instead with a **radial-out** burn ($\sim 7,100$ m/s), and compare the Δv the two cost for the same result.
3. Try a burn just *below* escape and one just *above* it, and compare — one stays a closed loop, the other opens and never returns.

What to look for: a prograde burn is far cheaper because it adds directly to your existing speed; a radial burn must supply the whole escape speed from scratch. Either way, once speed $\geq$ escape speed the eccentricity reaches 1.0 and the path becomes an open **hyperbola** — the object leaves and never returns. This open flyby geometry is exactly what "scattering" past a body looks like.

While you watch, think about:

- What value does the eccentricity readout cross at the exact moment the orbit stops closing?
- Once the path is a hyperbola, does the object ever come back to its starting point?
- How is this open curve related to what a comet does when it swings past a planet and leaves?

Check your understanding: When a satellite's speed reaches escape velocity, its eccentricity reaches 1.0 and its orbit...

- Becomes a tighter circle, since the extra energy pulls the path inward
- Opens from a closed ellipse into an unbound path — it leaves and never returns
- Stays exactly the same, because eccentricity does not affect the shape
- Stops abruptly, with the satellite losing its orbital motion entirely

Exercise 12 — Escaping is far cheaper prograde than radial

You just escaped Earth two different ways — and if you watched the fuel gauge, one of them was a bargain and the other nearly emptied the tank. Both reached the same open hyperbola. So why the enormous difference in cost?

It's the same lesson as the tangential-vs-radial burn, now with real stakes. A **prograde** burn adds directly on top of the speed you already have, so you only need to make up

the gap — about $0.41 \times$ your circular speed — to reach escape. A **radial** burn is aimed sideways to your motion; it can't borrow any of your existing speed, so it has to supply almost the *entire* escape speed from scratch. Same precious Δv , wildly different mileage. The rule of thumb writes itself: **when you can, burn along your velocity.**

Predict first: Before you watch the gauge: you'll escape once prograde and once radial-out. Which do you think will drain more of your Δv budget — and by a little, or by a lot?

Try it in the simulator:

1. Still in the **Escape & unbound** scenario, watch the Δv budget gauge as you escape with a **prograde** burn ($\sim 2,950$ m/s) and note how much fuel it took.
2. Hit **refuel / reset budget** and re-click the **Escape & unbound** scenario to get back on the circular orbit (refuel only refills the tank — it doesn't reset the orbit). Then escape with a **radial-out** burn ($\sim 7,100$ m/s) and compare the two gauge readings side by side for the identical outcome.

What to look for: the prograde escape barely dents the fuel gauge, while the radial escape drains most of the tank for the same result. Adding speed along the direction you're already moving is far more fuel-efficient than throwing it sideways.

While you watch, think about:

- Both burns escaped Earth — so why did one cost more than twice as much Δv as the other?
- Why can a prograde burn “borrow” your existing speed while a radial burn can't?
- How does this reinforce the rule you met earlier about which direction to point a burn?

Check your understanding: From a circular orbit, which takes far less Δv (fuel): escaping with a prograde burn, or a radial-outward burn?

- Radial is cheaper, because it points directly along the escape direction
- Prograde is much cheaper — it adds to the speed you already have
- They cost exactly the same, since escape depends only on total energy
- Neither can reach escape from a circular orbit without a second burn

Exercise 13 — Radial infall: cancel the sideways speed

You saw this idea as a thought experiment back in Part B — kill the sideways speed and you fall straight in. Now watch it play out for real, and from way up high, where the drop is spectacular.

Start far out at 20,000 km and fire a big retrograde burn to cancel your orbital speed completely. With nothing left carrying you sideways, gravity simply takes over and pulls you straight down. And because you fall through a huge distance, you keep **accelerating the whole way in**, arriving at the surface at tremendous speed. The higher you start the plunge, the faster you're moving when you arrive — all that height converts into speed on the way down.

Predict first: Before you burn: you're parked far out at 20,000 km and you cancel all your sideways speed. Do you think you'll stay parked out there, drift outward, or fall straight in — and if you fall, will you arrive slowly or fast?

Try it in the simulator:

Open the **Radial infall** scenario (starts far out at 20,000 km). Apply a big **retrograde** burn ($\sim 3,900$ m/s) to cancel the orbital speed entirely, and watch the descent and the speed readout all the way to impact.

What to look for: with the sideways speed gone, there's nothing to "keep missing" the Earth with. The object falls radially straight in, accelerating the whole way, and slams into the surface at high speed — the higher you start, the faster it arrives.

While you watch, think about:

- As the object falls, does the speed readout climb, hold steady, or drop? Why?
- This is the same "kill the sideways speed" idea from Part B — how does starting from way out at 20,000 km change how fast it arrives?
- Where did the enormous impact speed come from, if you only spent Δv to *slow down*?

Check your understanding: You are far out at 20,000 km and cancel your orbital (side-ways) speed to zero. What happens?

- You stay parked at 20,000 km, held there by the balance of forces
- You fall radially straight down, speeding up the whole way to impact
- You spiral slowly outward as the remaining energy redistributes itself
- You settle into a circular orbit at 20,000 km, just a slower one

PART E • Atmospheric drag & disposal

The one perturbation that ends missions — and can be used to dispose of them.

Exercise 14 — Watch an orbit decay and re-enter

Down in low orbit, space isn't perfectly empty. There's a wisp of upper atmosphere — thin, but not nothing — and a satellite plowing through it at 7.8 km/s feels a steady **drag**, exactly like the wind resistance on a car. That drag saps orbital energy little by little, shrinking the orbit until, eventually, the satellite spirals down and re-enters.

The crucial fact is *how* the air thickens as you descend: not gradually, but **exponentially**. Drop a hundred kilometers and you can be flying through many times more air. So altitude makes an enormous difference to survival time — a satellite at 400 km can last months, while one at 250 km may last only days. It's why the ISS at about 420 km needs a periodic reboost, or it too would come down.

Predict first: Before you speed up time: you'll watch orbits decay from three starting heights — 300 km, then 250 km, then 400 km. Rank them: which do you predict will spiral in fastest, and which will hang on longest?

Try it in the simulator:

1. Use the **Drag decay** scenario. Place a circular orbit at ~ 300 km and speed up time warp. Watch the spiral tighten and plunge.
2. Reset and try ~ 250 km, then ~ 400 km, comparing how long each one survives before re-entry.

What to look for: the orbit slowly shrinks into a tighter spiral, then plunges. Lower starting altitudes decay dramatically faster (250 km in days; 400 km in months). At re-entry the speed is ~ 7.8 km/s — about 17,500 mph.

While you watch, think about:

- Did the survival times scale gently with altitude, or did a small drop in height make a huge difference?
- Why does the air get so much thicker over just a few hundred kilometers of descent?
- Given all this, why does the ISS need regular reboosts to stay at ~ 420 km?

Check your understanding: How does the starting altitude affect how quickly a LEO orbit decays from drag?

- Higher orbits decay faster, because they sweep through more volume per lap
- Lower orbits decay far faster, because the air is exponentially denser down low
- All altitudes decay at much the same rate once drag is acting at all
- Altitude has no effect; decay depends only on the satellite's own shape

Exercise 15 — The drag paradox: friction speeds you up

Here's a genuine head-scratcher. Drag is a friction force — it opposes motion, it takes energy away. Every instinct says a satellite fighting through air should **slow down**. Watch the simulator, though, and you'll see the speed readout doing the opposite: as the orbit decays, the satellite gets **faster**.

The resolution is that pattern from Module 1: **a lower orbit is a faster orbit**. Drag steals orbital energy, which drops the satellite to a lower altitude — and down there, the speed required to stay in orbit is *higher*. So the satellite trades altitude for speed and ends up moving faster than before, even though friction is “slowing” it. The friction genuinely removed energy; it's just that lower orbits demand more speed, not less.

Predict first: Before you watch the readout: as drag drags a satellite down to a lower orbit, do you think its speed will drop (it is friction, after all), stay the same, or actually climb?

Try it in the simulator:

While watching the decay in the **Drag decay** scenario, keep an eye on the **speed** readout as the altitude falls, and note which way it moves as the orbit shrinks.

What to look for: even though drag is a friction force, the satellite's speed *increases* as it spirals down — because a lower orbit is a faster orbit.

While you watch, think about:

- The speed went *up* while friction was acting — so what exactly did the drag take away, if not speed?
- How does “lower orbit = faster orbit” from Module 1 resolve the apparent contradiction?
- Is the satellite really gaining energy overall, or is it trading one kind for another?

Check your understanding: As drag pulls a satellite to a lower orbit, its orbital speed...

- Decreases, because friction acts directly against the direction of travel

- Increases — drag drops it to a lower orbit, where circular speed is higher
- Stays the same, since drag removes energy without changing the speed
- Falls to zero quickly, once drag has cancelled the orbital motion

Exercise 16 — Disposal: burning up vs. boosting away

Every satellite eventually dies, and leaving a dead one in a busy orbit is a collision hazard for everyone else. So end-of-life **disposal** is part of the plan — and the smart way to do it depends entirely on where the satellite lives, because it comes down to Δv economics.

For a satellite in **LEO**, disposal is cheap. A small **retrograde** burn dips your perigee down into the thicker atmosphere, and from there drag does all the rest of the work for free — it drags you down to burn up on re-entry. For a satellite way out at **GEO**, there's no useful air to help, and deorbiting all the way to Earth would cost a prohibitive ~ 1.5 km/s of Δv you don't have to spare. So GEO satellites do the opposite: a tiny (~ 11 m/s) prograde nudge lifts them into a **"graveyard" orbit** just above the belt, safely out of everyone's way.

The point: dropping out of LEO takes little Δv — drag does the rest. Dropping from GEO all the way down would take ~ 1.5 km/s, so instead GEO satellites make a tiny (~ 11 m/s) boost to a "graveyard" orbit above the belt.

Predict first: Before you reason it out: a dead LEO satellite and a dead GEO satellite both need disposing of. Do you think you'd get rid of them the same way — and if not, which one is cheap to drop out of the sky and which one is cheaper to push *up* and away?

While you watch, think about:

- Why is drag a disposal *helper* in LEO but no help at all at GEO?
- Deorbiting from GEO costs ~ 1.5 km/s but a graveyard boost costs only ~ 11 m/s — why is "up and away" the sensible choice there?
- Why is it worth reserving a little fuel at the very end of a mission just for this final burn?

Check your understanding: Why are LEO satellites disposed of by re-entry, but GEO satellites are boosted to a "graveyard" orbit instead?

- GEO satellites are too massive to move once they reach their final slot
- From LEO a small burn lets drag finish the job; from GEO deorbiting costs far more
- There is no air near GEO to help, and controlled re-entry is not permitted there
- GEO satellites carry no propellant reserve, so they cannot be moved at all

PART F · Radiation pressure, HAMR & solar sails

The photon wind, and why some objects are nearly impossible to predict.

Exercise 17 — Radiation pressure and area-to-mass

Sunlight doesn't just light things up — it gently **pushes** on them. Light carries momentum, so a stream of photons hitting a satellite acts like an incredibly faint, steady wind. It's far too weak to matter down low where drag dominates, but above about 1,000 km, where the air is gone, this **radiation pressure** becomes a real perturbation that nudges orbits over time.

How hard an object gets shoved depends on its **area-to-mass ratio** — its illuminated area divided by its mass. The push comes from area (more surface catches more photons), but the resulting acceleration is that force divided by mass ($a = F/m$). So a dense, compact satellite barely budes, while a big, light, floppy object gets thrown around. It's the exact same area÷mass dependence that governs drag — light, spread-out things feel these forces the most.

Predict first: Before you move the slider: two objects sit at the same high altitude in the same sunlight — one a heavy, compact satellite, the other light and spread-out. Which do you predict gets pushed off course more by sunlight, and why?

Try it in the simulator:

1. Watch the area-to-mass effect where this simulator models it — in the air. (Plainly said: the sim models **drag only, below 1,000 km**; sunlight isn't simulated.) In the **Drag decay** scenario, place a low circular orbit (~300 km), let it decay for a while, then raise the **area-to-mass** slider up toward "HAMR debris" and watch the decay accelerate dramatically — same orbit, same air, more area per unit mass.
2. Now carry that logic up high: above ~1,000 km the air is gone, but sunlight is still pushing — and the same area ÷ mass ratio decides who gets shoved. The flimsy, high-area object whose orbit just decayed fastest is exactly the one the photon wind would perturb most.

What to look for: raising area-to-mass made the low orbit decay far faster — the acceleration from a surface force scales with area ÷ mass. Sunlight works by the same rule: it pushes on everything, and a big, light object (high area-to-mass) is shoved far more than a dense compact one. (This simulator models drag only, below 1,000 km; radiation pressure is the same logic applied up high.)

While you watch, think about:

- Both objects feel the same sunlight — so why does the lighter one move off course more?
- Where have you seen this same area÷mass dependence before in this module?
- Why does radiation pressure only start to matter up high, above about 1,000 km?

Check your understanding: Two objects at the same high altitude feel the same sunlight. Which one has its orbit perturbed more by radiation pressure?

- The heavier, compact one, since more mass means more momentum to redirect
- The lighter one, with more illuminated area per unit of mass
- Both are perturbed equally, because they receive identical sunlight
- Neither — radiation pressure is far too weak to affect an orbit at all

Exercise 18 — HAMR objects and solar sails

Now take that area-to-mass idea to its extreme. Picture the flimsiest junk in orbit: a scrap of foil, a torn thermal blanket, a floppy "trash-bag" of debris. These are **HAMR** objects — High Area-to-Mass Ratio — and sunlight blows them around like leaves in a breeze.

That makes them a nightmare to track, and for a subtle reason. It's not just that the push is strong; it's that a floppy, **tumbling** object keeps turning different amounts of

area toward the Sun. So the force on it is constantly changing in ways that are nearly impossible to model — its orbit wanders unpredictably. The flip side is beautiful: harness that same steady photon push on purpose, with a big reflective sheet, and you get a **solar sail** — a spacecraft that “sails” on sunlight alone, carrying no propellant at all.

The point: a HAMR object’s trouble isn’t just a big push — it’s that its tumbling, changing area makes that push vary unpredictably, so its future orbit is hard to pin down.

[Figure/insert: see the online worksheet]

While you watch, think about:

- Why is a *tumbling* scrap of foil even harder to predict than a rigid one with the same area-to-mass?
- If the push varied predictably, would the orbit be so hard to forecast? What’s the real culprit?
- A solar sail and a piece of HAMR junk feel the very same force — so what makes one useful and the other a headache?

Check your understanding: Why are HAMR objects (like tumbling foil debris) so hard to predict?

- They are effectively invisible to radar, so no tracking data ever exists
- Sunlight pushes them hard, and tumbling keeps changing their exposed area
- They travel faster than any tracked object, outrunning the prediction window
- They are so light that gravity barely acts on them in the usual way

PART G · J2 & sun-synchronous orbits

Turning Earth’s bulge from a nuisance into a feature.

Exercise 19 — Earth’s bulge swivels the orbit plane

Every orbit so far has quietly assumed Earth is a perfect sphere. It isn’t. Earth spins, and that spin makes it bulge outward at the equator — it’s a few dozen kilometers fatter around the middle than pole-to-pole. That bulge means gravity doesn’t pull quite straight toward the center for a tilted orbit.

Instead, the extra pull from the bulge gives a tilted orbit a sideways **torque**. Rather than yanking the satellite down, this torque slowly **swivels the entire orbit plane** around Earth’s axis — the spot where the orbit crosses the equator (its node) creeps around, degree by degree, over days and weeks. This steady swiveling is the **J2 effect**, and for most satellites it’s the single biggest departure from a clean, ideal Keplerian orbit. (“J2” is just physicists’ label for the bulge term in Earth’s gravity.)

[Figure/insert: see the online worksheet]

Predict first: Before you run it: Earth’s equatorial bulge tugs on a tilted orbit. Do you think that tug mainly speeds the satellite up, rounds out its shape, or slowly rotates the whole orbit plane around Earth’s axis?

Try it in the simulator:

Recall that Earth is not a perfect sphere — it bulges at the equator. Read the note, then open the **Sun-synchronous** scenario and speed up time to watch the orbit plane precess (swivel) over days and weeks.

What to look for: the orbit plane itself slowly rotates around Earth's axis — the orbit isn't changing size or speed, its whole plane is swiveling.

While you watch, think about:

- What is actually changing as the orbit precesses — its size, its speed, or its orientation in space?
- Why does a lumpy, bulging Earth push the plane sideways rather than just pulling the satellite straight down?
- This perturbation is continuous but very *predictable* — how might a clever engineer turn that to their advantage?

Check your understanding: What does Earth's equatorial bulge (the J2 effect) do to a tilted orbit?

- It speeds the satellite up slightly on each equator crossing
- It slowly swivels the whole orbit plane around Earth's axis — the node precesses
- It rounds the orbit out, pulling eccentricity gradually toward zero
- It has no effect at all unless the satellite fires its thrusters

Exercise 20 — Sun-synchronous orbits: a perturbation as a feature

Most perturbations are nuisances to fight. J2 is the rare one engineers **welcome** — because if you can't beat the swivel, you can aim it. The trick is to make the orbit plane precess at exactly the right rate to stay locked to the Sun.

Earth goes once around the Sun per year, so the Sun's direction shifts by about **0.9856° per day**. J2's precession rate depends on the orbit's **altitude, shape, and inclination** together — so for a **given altitude** you pick the inclination that makes the plane precess by exactly that amount (near **98°**, a slightly-past-polar orbit, for typical low sun-sync orbits) — **one full turn per year**. Now your orbit keeps a constant angle to the Sun as the year goes by. That's a **sun-synchronous** orbit.

The payoff: the satellite crosses over every location at the **same local sun time** on every pass. Same sun angle, same shadow lengths, every single time. For imaging, weather, and change-detection that consistency is gold — you can lay two images side by side and trust that differences are real changes on the ground, not just the Sun having moved.

Predict first: Before you run the year: two sun-sync orbits (a cyan circular one and a gold elliptical one) are tilted near 98°. As a full year of J2 precession plays out, do you predict their planes will drift away from the Sun, or stay locked at the same angle to it?

Try it in the simulator:

1. In the **Sun-synchronous** scenario, speed up time and watch a full year go by. Two sun-sync orbits are drawn — a **circular** one (cyan) and an **elliptical** one (gold) — and the arriving-sunlight arrow shows the Sun's direction.
2. Watch whether both planes keep the same angle to the Sun as the year turns. (This schematic simply applies the sun-sync precession rate to the whole scene to show

the idea; in reality each altitude and shape needs its own slightly different inclination to precess at exactly that rate — that’s the knob the designer tunes.)

What to look for: if the plane swivels $0.9856^\circ/\text{day}$ (one full turn per year), the orbit keeps a constant orientation to the Sun. In this schematic that sun-sync rate is applied to the whole scene; for a real mission, the designer tunes the inclination ($\sim 98^\circ$ for typical low sun-sync orbits) to each orbit’s own altitude and shape so it precesses at exactly that rate. The satellite crosses every location at the same local sun time, so lighting and shadows are consistent every pass.

While you watch, think about:

- Both rings stayed Sun-locked here because the schematic precesses them together — for a real pair of orbits with different shapes, what would the designer have to adjust on each one to keep them both Sun-locked?
- Why is crossing each place at the same local sun time so valuable for comparing images taken weeks apart?
- This uses J2 as a feature instead of fighting it — what other “nuisance” forces might be turned into tools the same way?

Check your understanding: Why might a sun-synchronous orbit (constant local sun time) be advantageous?

- It lets the satellite hover over one chosen spot, the way GEO does
- Every pass images a place under the same sun angle and shadows
- It removes the need for station-keeping propellant altogether
- It keeps the satellite in permanent sunlight, so batteries are unnecessary

PART H • Custody & targeted observations

If orbits drift, you have to keep looking.

Exercise 21 — Keeping “custody” of an object

Here’s where this whole module lands. You’ve now seen a long list of forces — drag, J2, radiation pressure, thruster burns — that all keep nudging an orbit off its ideal ellipse. The practical consequence is blunt: a set of orbital elements is only good for a while. It goes stale. So tracking a satellite can’t be a one-and-done measurement.

It has to be an **ongoing job**. Operators re-observe an object again and again, updating its elements each time to keep pace with the drift. Maintaining that unbroken chain of “we still know which object this is and where it is right now” is called keeping **custody**. Break the chain — wait too long between looks, or mix up two objects that passed close together — and you may lose custody and have to **re-acquire**: search a patch of sky, find the object again, and sort out which track is which. Doing this across everything in orbit is a core task of **Space Domain Awareness (SDA)**.

The point: custody isn’t a fact you file away once — it’s a chain of fresh observations you have to keep renewing as the orbit drifts.

[Figure/insert: see the online worksheet]

While you watch, think about:

- Why can't you just measure an orbit once, store the elements, and be done with it?
- What are two ways you might *lose* custody of an object?
- Which perturbations from this module are the reason custody has to be actively maintained?

Check your understanding: What does it mean to maintain “custody” of a space object?

- Physically capturing the object with another spacecraft and holding it
- Keeping an unbroken, up-to-date track of which object it is and where
- Holding legal ownership of the satellite and its assigned orbital slot
- Storing one perfect TLE and reusing it for the life of the mission

Exercise 22 — Why observations must be targeted

A telescope or a radar can only stare at a **small patch of sky** at any one moment — think of peering through a soda straw. Meanwhile there are tens of thousands of objects up there to keep track of. You can't sweep the whole sky hoping to stumble across the one you want; you'd almost always be pointed at emptiness.

So observations are **targeted**. You take the object's known elements, use them to predict exactly *where* in the sky it should appear and *when*, and then slew the sensor to that spot at that moment. A good prediction turns an impossible “search everywhere” into a simple “look right here, now.” And this is exactly why custody matters: the prediction is only as good as the elements it's built from. Thanks to all the perturbations in this module, stale elements point the sensor at empty sky — you have to keep them fresh to keep hitting your target.

The point: you use the object's known (and freshly updated) elements to predict where it should be, then point the sensor there at the right moment. A good orbit prediction turns “search the whole sky” into “look right here, now.”

Predict first: Before you reason it out: a single telescope sees only a tiny sliver of sky, and there are tens of thousands of objects. Do you think it's better to sweep the sky at random, or to predict where a specific object will be and point straight there? What would you need in order to do the latter?

While you watch, think about:

- Why is randomly sweeping the sky a hopeless way to find one particular object?
- What piece of information lets you point the sensor at the right place *and* the right time?
- How does a stale set of elements — from the perturbations you studied — cause a sensor to miss its target entirely?

Check your understanding: Why are space-surveillance observations usually “targeted” rather than just staring at random sky?

- Sensors can watch the whole sky at once, so targeting is a formality
- A sensor sees a small patch, so the predicted orbit says where to point
- Objects stay invisible until you already know their catalogue name
- International law requires each observation to be filed against a target

Key points

- Real satellites feel forces beyond Earth's point-mass gravity: **atmospheric drag**, the **equatorial bulge (J2)**, **solar radiation pressure**, **outgassing**, and deliberate **thruster burns**. These push the orbit off its ideal Keplerian ellipse, so we lose the ability to predict trajectories far ahead.
- **Chemical thrusters** give brief, powerful impulses (predictable coasts in between); **ion thrusters** give gentle continuous thrust for weeks (efficient, but hard to predict — e.g. Starlink).
- **Delta-v** (a change in speed, in m/s or km/s) is the currency of maneuvers. A satellite carries a finite supply; when it runs out, the mission is effectively over.
- **Tangential (prograde/retrograde) burns change the orbit's energy efficiently** — they raise or lower the opposite side of the orbit. A **radial** burn does almost no work on the orbit, so it's a poor way to change altitude. To fight drag, burn **prograde**.
- Kill the tangential speed entirely and there's nothing holding you up sideways — the satellite **falls straight back toward Earth**, accelerating all the way in.
- A big orbit change is a **two-step macro-maneuver**: **(1)** burn to raise your apogee out to the desired distance; **(2)** at that apogee, change speed to match the **circular speed** there, which circularizes the orbit. Same recipe for any Hohmann-type transfer.
- A **GTO (geo-transfer orbit)** is the ellipse of step 1: low perigee, apogee at GEO. The step-2 "**goldilocks**" **prograde burn (~1.46 km/s)** at apogee circularizes it. LEO→GEO totals ~3.9 km/s.
- **Escape velocity** = $\sqrt{2} \times$ the local circular speed: **11.2 km/s from the surface** (~10 km/s from a 1,500 km orbit). Reach it and eccentricity hits 1.0 — the closed ellipse opens into an **unbound hyperbola** and the object leaves forever (the geometry of a flyby / scattering). Escaping **prograde** is far cheaper than escaping **radially**.
- **Atmospheric drag** shrinks a LEO orbit until re-entry; lower orbits decay far faster (400 km $\approx$ months, 250 km $\approx$ days). Objects re-enter at ~7.8 km/s (~17,500 mph). This is used to **dispose** of LEO satellites by burning them up.
- The drag "paradox", stated carefully because the short version misleads: **drag never pushes a satellite along**. It removes energy, the orbit shrinks, and circular speed is higher at lower altitude — so the satellite ends up moving *faster* while continuously losing energy. Friction did not accelerate it; the smaller orbit it fell into demands more speed.
- **Radiation pressure** and drag both scale with **area $\div$ mass**. **HAMR** objects (high area-to-mass — foil, insulation, bags) are flung around by sunlight and, being flimsy and tumbling, change their area unpredictably, so their orbits are notoriously hard to forecast. The same physics is the basis of **solar sails**.
- Earth's bulge (**J2**) torques the orbit plane so it swivels. Choosing an inclination (~98°) makes it swivel **0.9856°/day = one turn per year**: a **sun-synchronous** orbit that crosses each place at the same local sun time — ideal for consistent-lighting imagery.
- Because orbits drift, tracking is continuous: **custody** is the ongoing, up-to-date knowledge of an object's identity and orbit. Sensors see only a small patch of sky, so observations are **targeted** using the (fresh) predicted orbit; stale elements point the sensor at empty sky.

Final quiz

Answers are checked in the online version.

Q1. What is "delta-v," and why does it limit a mission?

- The satellite's distance from Earth, which shrinks steadily over the mission
- A change in speed you spend to maneuver — the supply is finite and never renewed
- The delay in radio signals travelling between the satellite and the ground
- The angle between the satellite's orbit plane and the Earth's equator

Q2. To efficiently raise a decaying orbit, which way should you burn?

- Tangentially, prograde — along the direction you are already moving
- Radially outward, straight away from Earth toward higher altitude
- Radially inward, using the resulting swing to carry you outward
- It makes no difference at all which way the thrust is pointed

Q3. What are the two steps of a standard orbit-raising (e.g. LEO→GEO) maneuver?

- Point at the target and thrust continuously the whole way there
- Kill all of your speed, then fall to the new altitude and settle
- Burn prograde to raise apogee, then at apogee match the speed there
- Burn radially outward once, then coast and let gravity do the rest

Q4. Roughly what is escape velocity from the surface of the Earth, and what happens when you reach it?

- About 1 km/s; the orbit tightens into a smaller, faster circle
- About 11 km/s; the closed ellipse opens and the object leaves for good
- About 100 km/s; the satellite loses its orbital motion and stops
- About 300,000 km/s; the orbit is unchanged by reaching that speed

Q5. A satellite loses energy to atmospheric drag. What happens to its speed, and why do TLEs go stale?

- It slows and eventually stops, and TLEs stay valid until it does
- Its speed is unchanged, and TLEs only go stale after a thruster burn
- It speeds up — the lower orbit is faster — and perturbations act continuously
- It speeds up, but the orbit stays perfectly predictable indefinitely

Q6. To deorbit a satellite from GEO so it falls to Earth, roughly what does it take — and how does it compare to reaching GEO in the first place?

- Two large burns totalling ~3,900 m/s, mirroring the whole trip up
- One retrograde burn of ~1,500 m/s at GEO, dropping perigee into the air
- No propellant at all, since the orbit decays on its own over time
- A radial burn aimed straight down at the Earth beneath the satellite

Learn more

- Orbital maneuver / delta-v — Wikipedia
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